

AERIS Expert Review Packet

Anomaly

Source / metric	openaq / PM10 (pm10)
Observed / expected baseline	72 / 28.4085 ug/m3
Baseline method	mean of the preceding 7 days of this series (Z-score detector)
Time	2026-08-04 16:00 UTC
Location	29.6700, -95.1285
Severity	severe
Detectors	zscore, stl, isolation_forest

Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
asos	Relative humidity (humidity)	percent	7 / 622	34.78 to 100; mean 73.2578	59.12 at 2026-08-04 15:55Z; 5 min before event
asos	Air temperature (temperature)	degC	7 / 622	23.8889 to 36.6667; mean 29.5777	32 at 2026-08-04 15:55Z; 5 min before event
asos	Wind direction (wind_direction)	degree	7 / 773	0 to 320; mean 143.389	260 at 2026-08-04 15:55Z; 5 min before event
asos	Wind speed (wind_speed)	m/s	7 / 810	0 to 33.4389; mean 2.7723	3.08666 at 2026-08-04 15:55Z; 5 min before event
noaa_gfs	500-hPa geopotential height (gh_500)	m	8 / 80	5879.26 to 5926.86; mean 5904.75	5915.87 at 2026-08-04 18:00Z; 2 h after event
noaa_gfs	Planetary boundary layer height (pbl_height)	m	8 / 80	11.2803 to 2456.55; mean 697.483	2075.04 at 2026-08-04 18:00Z; 2 h after event
noaa_gfs	Precipitable water (precipitable_water)	mm	8 / 80	29.2 to 58.478; mean 37.4041	38.1486 at 2026-08-04 18:00Z; 2 h after event
noaa_gfs	Surface pressure (surface_pressure)	hPa	8 / 80	1006.72 to 1015.12; mean 1010.69	1011.9 at 2026-08-04 18:00Z; 2 h after event
noaa_gfs	850-hPa temperature (t_850)	degC	8 / 80	18.2063 to 21.0207; mean 19.34	18.91 at 2026-08-04 18:00Z; 2 h after event
noaa_gfs	10-m eastward wind (u_10m)	m/s	8 / 80	-3.62395 to 2.00478; mean -0.0099	0.106125 at 2026-08-04 18:00Z; 2 h after event
noaa_gfs	10-m northward wind (v_10m)	m/s	8 / 80	0.378384 to 5.58564; mean 2.4917	2.36153 at 2026-08-04 18:00Z; 2 h after event
openaq	Ozone (ozone)	ppm	17 / 945	-0.004 to 0.078; mean 0.0225	0.026 at 2026-08-04 16:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
openaq	PM10 (pm10)	ug/m3	3 / 182	13 to 172; mean 51.7582	72 at 2026-08-04 16:00Z; at event time
openaq	PM2.5 (pm25)	ug/m3	72 / 2702	2.27 to 48.02; mean 11.542	23.8 at 2026-08-04 16:00Z; at event time
openweather	Cloud cover (cloud_cover)	percent	5 / 324	0 to 100; mean 13.6296	0 at 2026-08-04 16:00Z; at event time
openweather	Relative humidity (humidity)	percent	5 / 324	45 to 95; mean 73.892	70 at 2026-08-04 16:00Z; at event time
openweather	Precipitation rate (precipitation)	mm/h	5 / 324	0 to 0.39; mean 0.0065	0 at 2026-08-04 16:00Z; at event time
openweather	Surface pressure (pressure)	hPa	5 / 324	1009 to 1017; mean 1012.78	1014 at 2026-08-04 16:00Z; at event time
openweather	Air temperature (temperature)	degC	5 / 324	25.09 to 37.35; mean 30.4719	32.13 at 2026-08-04 16:00Z; at event time
openweather	Wind direction (wind_direction)	degree	5 / 324	0 to 325; mean 180.676	151 at 2026-08-04 16:00Z; at event time
openweather	Wind speed (wind_speed)	m/s	5 / 324	0.45 to 6.47; mean 2.1614	2.24 at 2026-08-04 16:00Z; at event time
purpleair	PM2.5 (pm25)	ug/m3	67 / 4164	0 to 557.7; mean 13.4716	16 at 2026-08-04 16:00Z; at event time
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-08-04 20:10Z; 4 h 10 min after event
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	count	3 / 3	1 to 1; mean 1	1 at 2026-08-04 20:10Z; 4 h 10 min after event
sentinel5p	CO column (s5p_co_column)	mol/m^2	5 / 5	0.0277646 to 0.0314233; mean 0.0303	0.0277646 at 2026-08-04 20:10Z; 4 h 10 min after event
sentinel5p	CO granules available (s5p_co_granule_available)	count	5 / 5	1 to 1; mean 1	1 at 2026-08-04 20:10Z; 4 h 10 min after event
sentinel5p	HCHO column (s5p_hcho_column)	mol/m^2	3 / 3	0.000121637 to 0.00030109; mean 0.0002	0.000121637 at 2026-08-04 20:10Z; 4 h 10 min after event
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	count	3 / 3	1 to 1; mean 1	1 at 2026-08-04 20:10Z; 4 h 10 min after event
sentinel5p	NO2 column (s5p_no2_column)	mol/m^2	3 / 3	3.75274e-05 to 5.22023e-05; mean 0	3.75274e-05 at 2026-08-04 20:10Z; 4 h 10 min after event
sentinel5p	NO2 granules available (s5p_no2_granule_available)	count	3 / 3	1 to 1; mean 1	1 at 2026-08-04 20:10Z; 4 h 10 min after event
sentinel5p	O3 granules available (s5p_o3_granule_available)	count	3 / 3	1 to 1; mean 1	1 at 2026-08-04 20:10Z; 4 h 10 min after event
sentinel5p	SO2 column (s5p_so2_column)	mol/m^2	3 / 3	-3.3714e-05 to 0.000178122; mean 0.0001	-3.3714e-05 at 2026-08-04 20:10Z; 4 h 10 min after event
sentinel5p	SO2 granules available (s5p_so2_granule_available)	count	3 / 3	1 to 1; mean 1	1 at 2026-08-04 20:10Z; 4 h 10 min after event
tceq	Carbon monoxide (co)	ppm	3 / 156	0.1 to 0.9; mean 0.2583	0.2 at 2026-08-04 16:00Z; at event time
tceq	Nitrogen dioxide (no2)	ppb	12 / 633	-2.7 to 33.6; mean 5.0397	1.2 at 2026-08-04 16:00Z; at event time

Source	Metric	Unit	Entities / points	Range; mean	Nearest to event
tceq	Sulfur dioxide (so2)	ppb	3 / 154	-0.5 to 7.2; mean -0.0682	0 at 2026-08-04 16:00Z; at event time

Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

Source	Metric	Values
asos	Relative humidity (humidity)	08-03 00Z 79.85, 06Z 88.24, 12Z 68.41, 18Z 46.44, 08-04 00Z 75.34, 06Z 90.78, 12Z 70.42, 18Z 53.5, 08-05 00Z 79.38, 06Z 92.15, 12Z 70.64, 18Z 47.4
asos	Air temperature (temperature)	08-03 00Z 28.09, 06Z 26.34, 12Z 30.64, 18Z 34.39, 08-04 00Z 28.59, 06Z 25.56, 12Z 30.45, 18Z 33.83, 08-05 00Z 28.82, 06Z 26.39, 12Z 30.36, 18Z 34.26
asos	Wind direction (wind_direction)	08-03 00Z 180.8, 06Z 135.1, 12Z 162.6, 18Z 151.9, 08-04 00Z 157.8, 06Z 70.82, 12Z 136.5, 18Z 163.2, 08-05 00Z 166.4, 06Z 93.84, 12Z 168.4, 18Z 199.4
asos	Wind speed (wind_speed)	08-03 00Z 2.387, 06Z 1.488, 12Z 2.275, 18Z 4.623, 08-04 00Z 3.374, 06Z 0.8376, 12Z 2.018, 18Z 5.535, 08-05 00Z 3.31, 06Z 1.232, 12Z 2.948, 18Z 3.887
noaa_gfs	500-hPa geopotential height (gh_500)	08-03 06Z 5887, 12Z 5880, 18Z 5896, 08-04 00Z 5900, 06Z 5914, 12Z 5900, 18Z 5916, 08-05 00Z 5918, 06Z 5926, 12Z 5911
noaa_gfs	Planetary boundary layer height (pbl_height)	08-03 06Z 167.6, 12Z 95.48, 18Z 2217, 08-04 00Z 938.1, 06Z 301.5, 12Z 101.9, 18Z 2014, 08-05 00Z 679.3, 06Z 323.4, 12Z 136.4
noaa_gfs	Precipitable water (precipitable_water)	08-03 06Z 31.42, 12Z 33.08, 18Z 34.9, 08-04 00Z 38.49, 06Z 33.44, 12Z 35.65, 18Z 39.53, 08-05 00Z 49.65, 06Z 38.7, 12Z 39.19
noaa_gfs	Surface pressure (surface_pressure)	08-03 06Z 1009, 12Z 1010, 18Z 1010, 08-04 00Z 1008, 06Z 1010, 12Z 1011, 18Z 1012, 08-05 00Z 1011, 06Z 1013, 12Z 1014
noaa_gfs	850-hPa temperature (t_850)	08-03 06Z 19.23, 12Z 18.47, 18Z 18.84, 08-04 00Z 19.88, 06Z 20, 12Z 19.02, 18Z 18.84, 08-05 00Z 19.2, 06Z 20.61, 12Z 19.3
noaa_gfs	10-m eastward wind (u_10m)	08-03 06Z 1.145, 12Z 1.115, 18Z -1.169, 08-04 00Z -2.435, 06Z 0.7441, 12Z 0.6063, 18Z -1.129, 08-05 00Z -0.2686, 06Z 1.071, 12Z 0.222
noaa_gfs	10-m northward wind (v_10m)	08-03 06Z 2.214, 12Z 0.8743, 18Z 1.766, 08-04 00Z 4.523, 06Z 2.888, 12Z 1.277, 18Z 2.549, 08-05 00Z 4.38, 06Z 2.752, 12Z 1.695
openaq	Ozone (ozone)	08-03 00Z 0.04612, 06Z 0.02463, 12Z 0.02048, 18Z 0.05143, 08-04 00Z 0.02236, 06Z 0.009273, 12Z 0.01327, 18Z 0.02896, 08-05 00Z 0.01676, 06Z 0.009621, 12Z 0.0106, 18Z 0.02456
openaq	PM10 (pm10)	08-03 00Z 22, 06Z 20.5, 12Z 50.18, 18Z 43.35, 08-04 00Z 41.67, 06Z 40.67, 12Z 73.56, 18Z 76.56, 08-05 00Z 56.13, 06Z 48.75, 12Z 72.87, 18Z 71.67
openaq	PM2.5 (pm25)	08-03 00Z 12.26, 06Z 9.387, 12Z 9.881, 18Z 12.12, 08-04 00Z 11.68, 06Z 15.27, 12Z 14.76, 18Z 11.79, 08-05 00Z 10.52, 06Z 11.23, 12Z 10.14, 18Z 8.753
openweather	Cloud cover (cloud_cover)	08-03 00Z 10.1, 06Z 1.069, 12Z 0, 18Z 2.1, 08-04 00Z 23.13, 06Z 1.8, 12Z 0.3333, 18Z 23, 08-05 00Z 79.68, 06Z 9.8, 12Z 0.1333, 18Z 2
openweather	Relative humidity (humidity)	08-03 00Z 76.3, 06Z 83.55, 12Z 75.97, 18Z 51.1, 08-04 00Z 71.7, 06Z 87.3, 12Z 77.63, 18Z 56.21, 08-05 00Z 75.52, 06Z 88.93, 12Z 78.93, 18Z 54.36
openweather	Precipitation rate (precipitation)	08-03 00Z 0, 06Z 0, 12Z 0, 18Z 0, 08-04 00Z 0, 06Z 0, 12Z 0.01, 18Z 0, 08-05 00Z 0.02387, 06Z 0.03533, 12Z 0, 18Z 0
openweather	Surface pressure (pressure)	08-03 00Z 1010, 06Z 1011, 12Z 1012, 18Z 1010, 08-04 00Z 1011, 06Z 1012, 12Z 1014, 18Z 1013, 08-05 00Z 1013, 06Z 1015, 12Z 1017, 18Z 1016
openweather	Air temperature (temperature)	08-03 00Z 29.36, 06Z 27.22, 12Z 30.23, 18Z 35.93, 08-04 00Z 30.17, 06Z 26.51, 12Z 29.98, 18Z 35.38, 08-05 00Z 30.17, 06Z 27.29, 12Z 29.94, 18Z 35.47
openweather	Wind direction (wind_direction)	08-03 00Z 195.4, 06Z 175.3, 12Z 215.5, 18Z 151.8, 08-04 00Z 161, 06Z 175, 12Z 192.8, 18Z 163.8, 08-05 00Z 180.6, 06Z 194, 12Z 190.8, 18Z 179.1
openweather	Wind speed (wind_speed)	08-03 00Z 2.316, 06Z 1.916, 12Z 1.322, 18Z 2.397, 08-04 00Z 2.572, 06Z 1.775, 12Z 1.748, 18Z 3.054, 08-05 00Z 3.095, 06Z 1.718, 12Z 1.86, 18Z 2.426

Source	Metric	Values
purpleair	PM2.5 (pm25)	08-03 00Z 13.86, 06Z 11.94, 12Z 12.27, 18Z 13.73, 08-04 00Z 13.27, 06Z 17.57, 12Z 15.78, 18Z 13.68, 08-05 00Z 14.44, 06Z 13.24, 12Z 11.02, 18Z 8.618
sentinel5p	Aerosol index granules (s5p_aer_ai_granule_available)	08-03 18Z 1, 08-04 18Z 1
sentinel5p	CH4 granules available (s5p_ch4_granule_available)	08-03 18Z 1, 08-04 18Z 1
sentinel5p	CO column (s5p_co_column)	08-03 18Z 0.03095, 08-04 18Z 0.02776
sentinel5p	CO granules available (s5p_co_granule_available)	08-03 18Z 1, 08-04 18Z 1
sentinel5p	HCHO column (s5p_hcho_column)	08-03 18Z 0.0002755, 08-04 18Z 0.0001216
sentinel5p	HCHO granules available (s5p_hcho_granule_available)	08-03 18Z 1, 08-04 18Z 1
sentinel5p	NO2 column (s5p_no2_column)	08-03 18Z 4.973e-05, 08-04 18Z 3.753e-05
sentinel5p	NO2 granules available (s5p_no2_granule_available)	08-03 18Z 1, 08-04 18Z 1
sentinel5p	O3 granules available (s5p_o3_granule_available)	08-03 18Z 1, 08-04 18Z 1
sentinel5p	SO2 column (s5p_so2_column)	08-03 18Z 0.0001056, 08-04 18Z -3.371e-05
sentinel5p	SO2 granules available (s5p_so2_granule_available)	08-03 18Z 1, 08-04 18Z 1
tceq	Carbon monoxide (co)	08-03 06Z 0.2389, 12Z 0.2722, 18Z 0.25, 08-04 00Z 0.2111, 06Z 0.2556, 12Z 0.3167, 18Z 0.2615, 08-05 00Z 0.3, 06Z 0.2222, 12Z 0.2722, 18Z 0.2
tceq	Nitrogen dioxide (no2)	08-03 06Z 8.049, 12Z 7.837, 18Z 4.965, 08-04 00Z 2.549, 06Z 4.78, 12Z 6.113, 18Z 3.278, 08-05 00Z 2.103, 06Z 3.751, 12Z 4.996, 18Z 2.381
tceq	Sulfur dioxide (so2)	08-03 06Z -0.2056, 12Z 0.7444, 18Z -0.07222, 08-04 00Z -0.04444, 06Z -0.2556, 12Z -0.16, 18Z -0.1077, 08-05 00Z -0.1, 06Z -0.2778, 12Z -0.2111, 18Z -0.2

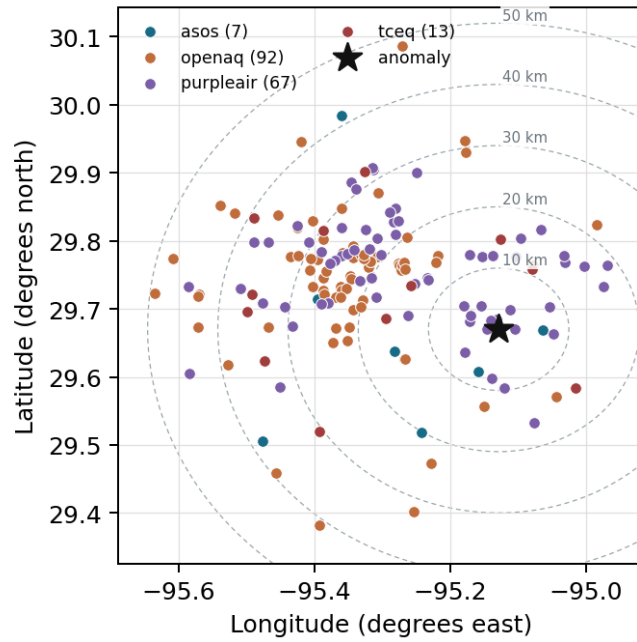
Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

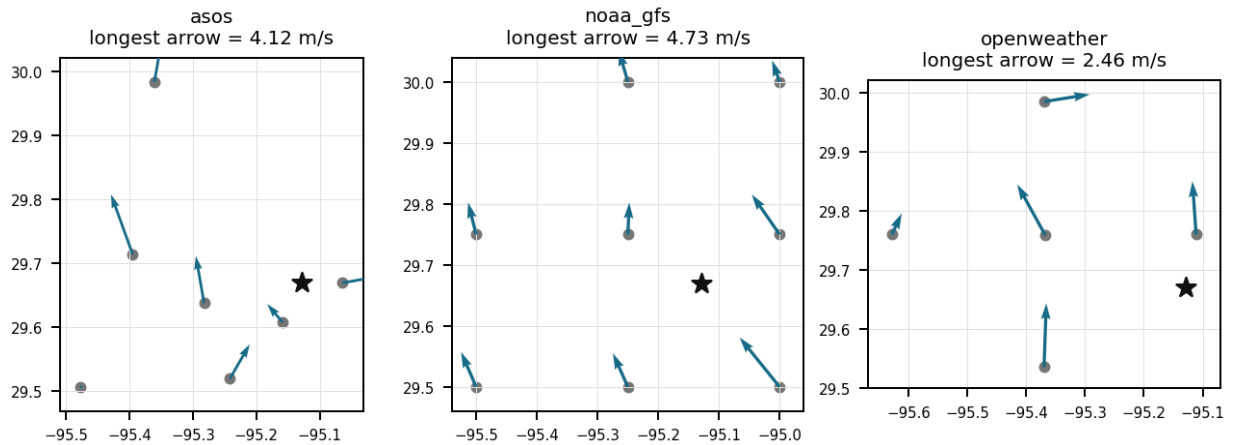
Source	Metric	Values
asos	Relative humidity (humidity)	T41 (6.224 km) 74.88, EFD (7.56 km) 73.96, HOU (15.304 km) 73.79, LVJ (20.054 km) 74.19, MCJ (26.202 km) 72.11, AXH (38.303 km) 72.74, IAH (41.517 km) 71.35
asos	Air temperature (temperature)	T41 (6.224 km) 29.5, EFD (7.56 km) 29.78, HOU (15.304 km) 29.83, LVJ (20.054 km) 29.44, MCJ (26.202 km) 29.83, AXH (38.303 km) 29.04, IAH (41.517 km) 30.02
asos	Wind direction (wind_direction)	T41 (6.224 km) 171.4, EFD (7.56 km) 136.5, HOU (15.304 km) 184.7, LVJ (20.054 km) 107.1, MCJ (26.202 km) 160.3, AXH (38.303 km) 70.56, IAH (41.517 km) 172.2
asos	Wind speed (wind_speed)	T41 (6.224 km) 3.168, EFD (7.56 km) 3.853, HOU (15.304 km) 3.447, LVJ (20.054 km) 2.042, MCJ (26.202 km) 3.715, AXH (38.303 km) 0.9667, IAH (41.517 km) 2.819
noaa_gfs	500-hPa geopotential height (gh_500)	gfs:29.75,-95.25 (14.723 km) 5905, gfs:29.75,-95.00 (15.268 km) 5905, gfs:29.50,-95.25 (22.259 km) 5904, gfs:29.50,-95.00 (22.624 km) 5904, gfs:29.75,-95.50 (36.964 km) 5905, gfs:30.00,-95.25 (38.518 km) 5905, gfs:30.00,-95.00 (38.729 km) 5905, gfs:29.50,-95.50 (40.594 km) 5905
noaa_gfs	Planetary boundary layer height (pbl_height)	gfs:29.75,-95.25 (14.723 km) 862.2, gfs:29.75,-95.00 (15.268 km) 741.7, gfs:29.50,-95.25 (22.259 km) 606.3, gfs:29.50,-95.00 (22.624 km) 650.1, gfs:29.75,-95.50 (36.964 km) 833.1, gfs:30.00,-95.25 (38.518 km) 767.6, gfs:30.00,-95.00 (38.729 km) 474.4, gfs:29.50,-95.50 (40.594 km) 644.5
noaa_gfs	Precipitable water (precipitable_water)	gfs:29.75,-95.25 (14.723 km) 37.6, gfs:29.75,-95.00 (15.268 km) 36.78, gfs:29.50,-95.25 (22.259 km) 38.49, gfs:29.50,-95.00 (22.624 km) 37.1, gfs:29.75,-95.50 (36.964 km) 37.44, gfs:30.00,-95.25 (38.518 km) 37.39, gfs:30.00,-95.00 (38.729 km) 36.18, gfs:29.50,-95.50 (40.594 km) 38.24

Source	Metric	Values
noaa_gfs	Surface pressure (surface_pressure)	gfs:29.75,-95.25 (14.723 km) 1011, gfs:29.75,-95.00 (15.268 km) 1012, gfs:29.50,-95.25 (22.259 km) 1011, gfs:29.50,-95.00 (22.624 km) 1012, gfs:29.75,-95.50 (36.964 km) 1010, gfs:30.00,-95.25 (38.518 km) 1009, gfs:30.00,-95.00 (38.729 km) 1010, gfs:29.50,-95.50 (40.594 km) 1010
noaa_gfs	850-hPa temperature (t_850)	gfs:29.75,-95.25 (14.723 km) 19.29, gfs:29.75,-95.00 (15.268 km) 19.25, gfs:29.50,-95.25 (22.259 km) 19.34, gfs:29.50,-95.00 (22.624 km) 19.31, gfs:29.75,-95.50 (36.964 km) 19.49, gfs:30.00,-95.25 (38.518 km) 19.34, gfs:30.00,-95.00 (38.729 km) 19.21, gfs:29.50,-95.50 (40.594 km) 19.49
noaa_gfs	10-m eastward wind (u_10m)	gfs:29.75,-95.25 (14.723 km) 0.3147, gfs:29.75,-95.00 (15.268 km) -0.1523, gfs:29.50,-95.25 (22.259 km) -0.02427, gfs:29.50,-95.00 (22.624 km) 0.01673, gfs:29.75,-95.50 (36.964 km) 0.1557, gfs:30.00,-95.25 (38.518 km) -0.3533, gfs:30.00,-95.00 (38.729 km) 0.1207, gfs:29.50,-95.50 (40.594 km) -0.1573
noaa_gfs	10-m northward wind (v_10m)	gfs:29.75,-95.25 (14.723 km) 2.439, gfs:29.75,-95.00 (15.268 km) 2.434, gfs:29.50,-95.25 (22.259 km) 2.547, gfs:29.50,-95.00 (22.624 km) 2.933, gfs:29.75,-95.50 (36.964 km) 2.758, gfs:30.00,-95.25 (38.518 km) 2.078, gfs:30.00,-95.00 (38.729 km) 2.248, gfs:29.50,-95.50 (40.594 km) 2.498
openaq	Ozone (ozone)	332 (0.0 km) 0.02433, 2800 (11.568 km) 0.02257, 2039 (14.072 km) 0.02162, 320 (14.284 km) 0.01722, 3378 (14.341 km) 0.02224, 5077791 (14.578 km) 0.01758, 3214 (14.733 km) 0.024, 325 (16.161 km) 0.006146, +9 more
openaq	PM10 (pm10)	271 (0.0 km) 45.83, 13380082 (14.341 km) 75.25, 15852419 (21.343 km) 34.29
openaq	PM2.5 (pm25)	268 (0.0 km) 17.22, 13955815 (12.759 km) 9.707, 14404954 (13.75 km) 11.56, 14157975 (14.287 km) 14.21, 3379 (14.341 km) 16.23, 15539682 (14.345 km) 12.09, 8967123 (14.399 km) 13.8, 8967479 (14.399 km) 12.48, +64 more
openweather	Cloud cover (cloud_cover)	grid:east (10.189 km) 10.26, grid:center (25.125 km) 13.78, grid:south (27.694 km) 14.48, grid:north (42.053 km) 14.4, grid:west (49.32 km) 15.23
openweather	Relative humidity (humidity)	grid:east (10.189 km) 77.4, grid:center (25.125 km) 72.56, grid:south (27.694 km) 74.15, grid:north (42.053 km) 71.88, grid:west (49.32 km) 73.45
openweather	Precipitation rate (precipitation)	grid:east (10.189 km) 0.003692, grid:center (25.125 km) 0.01109, grid:south (27.694 km) 0.006, grid:north (42.053 km) 0.01169, grid:west (49.32 km) 0
openweather	Surface pressure (pressure)	grid:east (10.189 km) 1013, grid:center (25.125 km) 1013, grid:south (27.694 km) 1013, grid:north (42.053 km) 1013, grid:west (49.32 km) 1013
openweather	Air temperature (temperature)	grid:east (10.189 km) 30.21, grid:center (25.125 km) 31.01, grid:south (27.694 km) 30.17, grid:north (42.053 km) 30.51, grid:west (49.32 km) 30.48
openweather	Wind direction (wind_direction)	grid:east (10.189 km) 182.6, grid:center (25.125 km) 198.1, grid:south (27.694 km) 185.5, grid:north (42.053 km) 152.2, grid:west (49.32 km) 185.3
openweather	Wind speed (wind_speed)	grid:east (10.189 km) 3.226, grid:center (25.125 km) 2.143, grid:south (27.694 km) 1.925, grid:north (42.053 km) 2.003, grid:west (49.32 km) 1.51
purpleair	PM2.5 (pm25)	128007 (1.624 km) 6.748, 161689 (1.695 km) 13.67, 222589 (1.835 km) 14.42, 268165 (2.295 km) 15.43, 222593 (3.625 km) 13.63, 186263 (4.256 km) 15.63, 98633 (4.617 km) 6.592, 244939 (4.67 km) 10.97, +59 more
tceq	Carbon monoxide (co)	482011039 (0.032 km) 0.2037, 482011035 (14.34 km) 0.1638, 482011052 (29.74 km) 0.3927
tceq	Nitrogen dioxide (no2)	482011039 (0.032 km) 2.685, 482011015 (10.969 km) 5.864, 482011035 (14.34 km) 9.982, 482011050 (14.586 km) 1.198, 482010026 (14.755 km) 7.555, 482010416 (16.16 km) 4.535, 482011052 (29.74 km) 15.79, 480391004 (30.466 km) 2.44, +4 more
tceq	Sulfur dioxide (so2)	482011039 (0.032 km) 0.3111, 482011035 (14.34 km) -0.3545, 482010051 (33.811 km) -0.2089

Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly



Event-nearest wind vectors from stored values (arrow points downwind)



Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

Label definitions, from the labeling guide

Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it.

Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold.

Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging.

Missing or thin evidence means Unsure, not Invalid.

A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it.

Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict.

Claim 1 of 36

The temperature on August 4th, 2026 was relatively high, with a mean value of 30.47°C, which may have contributed to the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note (optional):

Claim 2 of 36

There is no evidence to suggest that any other factors such as wind direction or speed, precipitation, or atmospheric pressure played a significant role in the air quality anomaly.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note (optional):

Claim 3 of 36

Industrial bulk material handling, earthmoving, or construction operations in the Deer Park and Houston Ship Channel industrial sector generated localized coarse particulate emissions observed by openaq and tceq.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 4 of 36

The openaq PM10 measurement at coordinates 29.670, -95.129 was 72.0 ug/m3 at 2026-08-04T16:00:00+00:00, compared with an expected value of 28.41 ug/m3 and a z-score of 4.36, and the anomaly was labeled severe.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 5 of 36

Longer-range transported dust is a secondary hypothesis because the synoptic pattern was hot and mostly cloud-free, but the available satellite trace-gas columns did not show a corresponding combustion signature and the evidence provided does not specifically demonstrate a regional dust plume over Houston at the anomaly time.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 6 of 36

OpenAQ measurements at the anomaly site recorded a PM10 concentration of 72.0 ug/m3 alongside a concurrent PM2.5 level of 23.8 ug/m3, demonstrating that coarse fraction particles comprised most of the particulate mass.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note (optional):

Claim 7 of 36

The air quality anomaly is related to PM2.5, which is elevated.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note (optional):

Claim 8 of 36

The air quality anomaly in Houston, Texas is likely caused by a combination of factors including high levels of PM2.5 and NO2 emissions from industrial activities and vehicle traffic.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note (optional):

Claim 9 of 36

The anomaly occurred on August 4th, around 16:00 UTC.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note (optional):

Claim 10 of 36

Local mechanically generated dust or resuspended road or construction material is supported because PM10 at the anomaly site reached 72.0 ug/m3 while co-located PM2.5 was only 23.8 ug/m3 and nearby PurpleAir PM2.5 was about 16.0 ug/m3, a pattern more consistent with coarse particles than with a fine-particle combustion plume.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note (optional):

Claim 11 of 36

The Houston anomaly occurred under hot, sunny daytime conditions favorable for dust resuspension and vertical mixing because asos and openweather showed temperatures near 32 C, humidity near 59 to 70 percent, cloud cover near 0 percent, and gfs indicated afternoon planetary boundary layer height near 2075 m.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note (optional):

Claim 12 of 36

Regional advection of coarse crustal aerosols across coastal Texas contributed to elevated baseline coarse particulate matter concentrations measured by openaq and purpleair.

Mark exactly one:	☐ V	☐ I	☐ U
-------------------	----------------------------	----------------------------	----------------------------

Note (optional):

Claim 13 of 36

Resuspended local dust is a leading hypothesis because PM₁₀ rose strongly while nearby PM_{2.5} and co-located combustion tracers remained much lower, a pattern that is more consistent with coarse particles from roads, bare soil, or disturbed surfaces than with smoke or fresh exhaust.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 14 of 36

The PM₁₀ anomaly recorded in Houston at 16:00 UTC on August 4, 2026, was dominated by coarse fraction particulate matter while PM_{2.5} levels remained comparatively low.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 15 of 36

An elevated PM₁₀ concentration of 72.0 ug/m³ was recorded on August 4, 2026, at 16:00 UTC at coordinates (29.670, -95.129), representing a severe anomaly with a z-score of 4.36 relative to an expected baseline of 28.41 ug/m³.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 16 of 36

The temperature was unusually high on August 4th, leading to increased emissions from industrial sources and vehicles.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 17 of 36

Localized industrial, port, construction, or unpaved-road material handling is a plausible cause because the PM10 enhancement was strong in a sparse PM10 network, the anomaly site had only light-to-moderate winds during a hot daytime period, and those conditions can produce short-lived coarse-particle plumes near sources.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 18 of 36

The PM2.5 levels were higher than usual in the surrounding area, potentially due to increased particulate matter emissions from local sources or distant wildfires.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 19 of 36

The air quality anomaly in Houston, Texas is caused by high levels of NO₂.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 20 of 36

The anomaly time and location were 2026-08-04T16:00:00+00:00 at coordinates 29.670, -95.129 in the Houston, Texas area.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 21 of 36

Local mechanically generated dust or resuspended road or construction material is further supported because the anomaly occurred during hot daytime conditions with low cloud cover and only modest winds, with about 32 C temperature, about 59 percent relative humidity at the nearest ASOS station, 0 percent cloud cover near the event, and wind speeds near 2 to 3 m/s, which are conditions that can dry surfaces and allow localized dust emissions to persist.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 22 of 36

OpenAQ PM10 monitoring in the area recorded 6-hour mean concentrations peaking at 73.56 ug/m3 at 12:00 UTC and 76.56 ug/m3 at 18:00 UTC on August 4, 2026.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 23 of 36

TCEQ monitoring at the nearest station recorded minimal co-pollutant concentrations of 1.2 ppb NO2 and 0.0 ppb SO2 at the time of the event, indicating that localized industrial combustion plumes were unlikely primary drivers.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 24 of 36

Localized wind-blown dust resuspension from unpaved surfaces and dry surface soils driven by daytime heating and surface winds contributed to the PM10 spike recorded by openaq.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 25 of 36

The location of the anomaly is near sensor 128007 (1.624 km) and monitor 482011039 (0.032 km).

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 26 of 36

A strong wind pattern on August 5th may have dispersed pollutants from nearby refineries and industrial areas towards the anomaly location.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 27 of 36

Regional transported smoke or a chemically rich industrial combustion plume is contradicted because ozone, CO, NO₂, and SO₂ were not strongly elevated near the anomaly time and same-day satellite columns for CO, NO₂, HCHO, and SO₂ were not enhanced, although transported mineral dust remains a weaker possibility because PM₁₀ stayed elevated across multiple 6-hour periods and southerly to southeasterly low-level flow was present.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 28 of 36

Low concurrent concentrations of nitrogen dioxide, sulfur dioxide, and carbon monoxide rule out major industrial combustion or fire plumes as the source of the PM₁₀ spike.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 29 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 30 of 36

At the time of the PM10 anomaly on August 4, 2026, at 16:00 UTC, the nearest PM2.5 sensor recorded a concentration of 23.8 ug/m3, demonstrating that coarse particulate matter composed the majority of the elevated PM10 mass.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 31 of 36

The Houston anomaly is unlikely to be primarily a combustion plume because nearby gaseous copollutants were not elevated at the anomaly time, with tceq showing CO at 0.2 ppm, NO2 at 1.2 ppb, and SO2 at 0.0 ppb near the site.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 32 of 36

At 2026-08-04T16:00:00+00:00 at the anomaly location, openaq PM2.5 was 23.8 ug/m3, openaq ozone was 0.026 ppm, tceq CO was 0.2 ppm, tceq NO2 was 1.2 ppb, and tceq SO2 was 0.0 ppb.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 33 of 36

The air quality anomaly in Houston, Texas is caused by high levels of CO.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 34 of 36

Daytime planetary boundary layer height exceeding 2000 meters accompanied by warm temperatures and low cloud cover facilitated localized surface dust resuspension.

Mark exactly one:☐ V☐ I☐ U**Note (optional):**

Claim 35 of 36

OpenAQ network data showed broad PM10 elevation across the region with 6-hour mean values reaching 73.56 ug/m3 to 76.56 ug/m3, supporting widespread coarse aerosol advection or dust resuspension rather than a single isolated point source.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Claim 36 of 36

The Houston anomaly is most consistent with a local mechanically generated coarse-particle source because openaq recorded PM10 at 72 ug/m3 against an expected 28.4 ug/m3 at 2026-08-04T16:00Z while co-located PM2.5 was only 23.8 ug/m3.

Mark exactly one:

☐ V

☐ I

☐ U

Note (optional):

Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.

